Because every common multiple of a and b is in $\langle a\rangle \cap \langle b\rangle = \langle c\rangle$, we see that every common multiple is of the form dc, that is, every common multiple is a multiple of c. Thus c is an lcm of a and b.

21. The subgroup of $\langle \mathbb{Z}, +\rangle$ generated by two integers $r, s \in \mathbb{Z}$ is $H = \{mr + ns \mid n, m \in \mathbb{Z}\}$. Now $H = \mathbb{Z}$ if and only if $1 \in H$, so $H = \mathbb{Z}$ if and only if $1 = mr + ns$ for some $m, n \in \mathbb{Z}$. If r and s are relatively prime, then 1 is a gcd of r and s, and the Euclidean algorithm together with the last statement in Theorem 46.9, show that 1 can be expressed in the form $1 = mr + ns$ for some $m, n \in \mathbb{Z}$. Conversely, if $1 = mr + ns$, then every integer dividing both m and n divides the righthand side of this equation and thus divides 1, so 1 is a gcd of r and s.

22. If a and n are relatively prime, then a gcd of a and n is 1. By Theorem 46.9, we can express 1 in the form $1 = m_1a + m_2n$ for some $m_1, m_2 \in \mathbb{Z}$. Multiplying by b, we get $b = a(m_1b) + (bm_2)n$. Thus $x = m_1b$ is a solution of $ax \equiv b \pmod{n}$.

23. Suppose that the positive gcd d of a and n in $\mathbb{Z}$ divides b. By Theorem 46.9, we can express d in the form $d = m_1a + m_2n$ for some $m_1, m_2 \in \mathbb{Z}$. Multiplying by b/d, we obtain $b = a(m_1b/d) + (bm_2/d)n$. Thus $x = m_1b/d$ is a solution of $ax \equiv b \pmod{n}$. Conversely, suppose that $ac \equiv b \pmod{n}$ so that n divides $ac - b$, say $ac - b = nq$. Then $b = ac - nq$. Because the positive gcd d of a and n divides the righthand side of $b = ac - nq$, it must be that d divides b also.

In $\mathbb{Z}_n$, this result has the following interpretation: $ax = b$ has a solution in $\mathbb{Z}_n$ for nonzero $a, b \in \mathbb{Z}_n$ if and only if the positive gcd of a and n in $\mathbb{Z}$ divides b.

24. *Step 1.* Use the Euclidean algorithm to find the positive gcd d of a and n.

Step 2. Use the technique of Exercise 6 to express d in the form $d = m_1a + m_2n$.

Step 3. A solution of $ax \equiv b \pmod{n}$ is $x = m_1b/d$.

We now illustrate with $22x \equiv 18 \pmod{42}$.

Step 1. Find the gcd of 22 and 42:

$$\begin{aligned} 42 &= 1(22) + 20, \\ 22 &= 1(20) + 2, \\ 20 &= 10(2), \end{aligned}$$

so 2 is a gcd of 22 and 42.

Step 2. Express 2 in the form $m_1(22) + m_2(42)$:

$$\begin{aligned} 2 &= 22 - 1(20) \quad \text{but} \quad 20 = 42 - 1(22), \\ 2 &= 22 - 1[42 - 1(22)] = 2(22) + (-1)(42), \end{aligned}$$

so $m_1 = 2$.

Step 3. A solution of $22x \equiv 18 \pmod{42}$ is $x = m_1b/d = (2)(18)/2 = 18$.

47. Gaussian Integers and Multiplicative Norms

1. Example 47.8 showed that $5 = (1 + 2i)(1 - 2i)$ is a factorization of 5 into irreducibles. Because $\mathbb{Z}[i]$ is a UFD, this factorization of 5 is unique up to unit factors. For example, $[i(1 + 2i)][-i(1 - 2i)] = (-2 + i)(-2 - i)$ is another factorization of 5 into irreducibles.

2. If α is a factor of 7 in $\mathbb{Z}[i]$, then $N(\alpha)$ must divide $N(7) = 49$, so $N(\alpha)$ must be 1, 7, or 49. If $N(\alpha) = 1$, then α is a unit and if $N(\alpha) = 49$, then the other factor must be a unit, and we are not interested in these cases. Thus we must have $N(\alpha) = 7$ if α is an irreducible dividing 7 and 7 is not irreducible. Because the equation $a^2 + b^2 = 7$ has no solutions in integers, it must be that 7 is already irreducible in $\mathbb{Z}[i]$.

3. Proceeding as in the answer to Exercise 2, if α is an irreducible factor of $4 + 3i$, then $N(\alpha)$ must be a divisor of $4^2 + 3^2 = 25$, and 5 is the only possibility if $4 + 3i$ is not irreducible. Thus we must have $\alpha = a + bi$ where a is ± 1 and b is ± 2 or where a is ± 2 and b is ± 1. A bit of trial and errror shows that $4 + 3i = (1 + 2i)(2 - i)$, and $1 + 2i$ and $2 - i$ are irreducible because they have the prime 5 as norm.

4. Proceeding as in the answer to Exercise 2, if α is an irreducible factor of $6 - 7i$, then $N(\alpha)$ must be a divisor of $6^2 + 7^2 = 85$, so $N(\alpha)$ must be either 5 or 17 if $6 - 7i$ is not irreducible. If $N(\alpha) = 5$, then $\alpha = a + bi$ where $a^2 + b^2 = 5$, so $a = \pm 1$ and $b = \pm 2$, or $a = \pm 2$ and $b = \pm 1$. We compute

$$\frac{6 - 7i}{1 + 2i} = \frac{6 - 7i}{1 + 2i} \cdot \frac{1 - 2i}{1 - 2i} = \frac{-8 - 19i}{5}$$

and find that the answer is not in $\mathbb{Z}[i]$. There is no use trying $\pm 1(1 + 2i)$ or $\pm i(1 + 2i)$, so we try $1 - 2i$. We obtain

$$\frac{6 - 7i}{1 - 2i} = \frac{6 - 7i}{1 - 2i} \cdot \frac{1 + 2i}{1 + 2i} = \frac{20 + 5i}{5} = 4 + i.$$

Thus $6 - 7i = (1 - 2i)(4 + i)$, and $1 - 2i$ and $4 + i$ are irreducibles because their norms are the primes 5 and 17 respectively.

5. We have $6 = 2 \cdot 3 = (-1 + \sqrt{-5})(-1 - \sqrt{-5})$. The numbers 2 and 3 are both irreducible in $\mathbb{Z}[\sqrt{-5}]$ because the equations $a^2 + 5b^2 = 2$ and $a^2 + 5b^2 = 3$ have no solutions in integers. (See Example 47.9 in the text.) If $-1 + \sqrt{-5}$ were not irreducible, then it would be a product $\alpha\beta$ where neither α nor β is a unit and $N(\alpha\beta) = N(\alpha)N(\beta) = 6$. This means that we would have to have $N(\alpha) = 2$ or $N(\alpha) = 3$, which is impossible as we have just seen. Because the only units in $\mathbb{Z}[\sqrt{-5}]$ are ± 1, we have two essentially different factorizations of 6.

6. We compute $\alpha\beta$:

$$\frac{7 + 2i}{3 - 4i} = \frac{7 + 2i}{3 - 4i} \cdot \frac{3 + 4i}{3 + 4i} = \frac{13 + 34i}{25}.$$

Now we take the integer 1 closest to 13/25 and the integer 1 closest to 34/25, and let $\sigma = 1 + 1i = 1 + i$. Then we compute

$$\begin{aligned} \rho = \alpha - \beta\sigma &= (7 + 2i) - (3 - 4i)(1 + i) \\ &= (7 + 2i) - (7 - i) = 3i. \end{aligned}$$

Then $\alpha = \sigma\beta + \rho$ where $N(\rho) = N(3i) = 9 < 25 = N(3 - 4i) = N(\beta)$.

7. We let $\alpha = 5 - 15i$ and $\beta = 8 + 6i$, and compute σ and ρ as in the answer to Exercise 6:

$$\frac{5 - 15i}{8 + 6i} = \frac{5 - 15i}{8 + 6i} \cdot \frac{8 - 6i}{8 - 6i} = \frac{-50 - 150i}{100}.$$

We take $\sigma = -i$, and $\rho = \alpha - \sigma\beta = (5 - 15i) - (-i)(8 + 6i) = -1 - 7i$, so

$$5 - 15i = (8 + 6i)(-i) + (-1 - 7i).$$

Continuing the Euclidean algorithm, we now take $\alpha = 8+6i$ and $\beta = -1-7i$, and obtain

$$\frac{8+6i}{-1-7i} = \frac{8+6i}{-1-7i} \cdot \frac{-1+7i}{-1+7i} = \frac{-50+50i}{50} = -1+i.$$

Because $-1+i \in \mathbb{Z}[i]$, we are done, and a gcd of $5-15i$ and $8+6i$ is $-1-7i$. Of course, the other gcd's are obtained by multiplying by the units -1, $\pm i$, so $1+7i, -7+i$, and $7-i$ are also acceptable answers.

8. T T T F T T T F T T

9. Suppose that $\pi = \alpha\beta$. Then $N(\pi) = N(\alpha)N(\beta)$. Because $|N(\pi)|$ is the minimal norm > 1, one of $|N(\alpha)|$ and $|N(\beta)|$ must be $|N(\pi)|$ and the other must be ± 1. Thus either α or β has norm ± 1, and is thus a unit by hypothesis. Therefore π is an irreducible in D.

10. a. We know that in $\mathbb{Z}[i]$, the units are precisely the elements $\pm 1, \pm i$ of norm 1. By Theorem 47.7, every element of $\mathbb{Z}[i]$ having as norm a prime in $\mathbb{Z}$ is an irreducible. Because $N(1+i) = 1^2+1^2 = 2$, we see that $1+i$ is an irreducible. The equation $2 = -i(1+i)^2$ thus gives the desired factorization of 2.

b. Every odd prime in $\mathbb{Z}$ is congruent to either 1 or 3 modulo 4. If $p \equiv 1 \pmod 4$, then Theorem 47.10 shows that $p = a^2+b^2 = (a+ib)(a-ib)$ where neither $a+ib$ nor $a-ib$ is a unit because they each have norm $a^2+b^2 = p > 1$, so p is not an irreducible.

Conversely, if p is not an irreducible, then $p = (a+ib)(c+di)$ in $\mathbb{Z}[i]$ where neither factor is a unit, so that both $a+bi$ and $c+di$ have norm greater than 1. Taking the norm of both sides of the equation, we obtain $p^2 = (a^2+b^2)(c^2+d^2)$, so we must have $p = a^2+b^2 = c^2+d^2$. Theorem 47.10 then shows that we must have $p \equiv 1 \pmod 4$.

We have shown that an odd prime p is not irreducible if and only if $p \equiv 1 \pmod 4$ so an odd prime p is irreducible if and only if $p \equiv 3 \pmod 4$.

11. *Property 1:* Let $\alpha = a+bi$. Then $N(\alpha) = a^2+b^2$. As a sum of squares, $a^2+b^2 \geq 0$.

Property 2: Continuing the argument for Property 1, we see that $a^2+b^2 = 0$ if and only if $a = b = 0$, so $N(\alpha) = 0$ if and only if $\alpha = 0$.

Property 3: Let $\beta = c+di$. Then $\alpha\beta = (a+bi)(c+di) = (ac-bd)+(ad+bc)i$, so

$$\begin{aligned} N(\alpha\beta) &= (ac-bd)^2+(ad+bc)^2 \\ &= a^2c^2-2abcd+b^2d^2+a^2d^2+2abcd+b^2c^2 \\ &= a^2c^2+b^2d^2+a^2d^2+b^2c^2 \\ &= (a^2+b^2)(c^2+d^2) = N(\alpha)N(\beta). \end{aligned}$$

[Of course it also follows from the fact that $|\alpha\beta|^2 = |\alpha|^2|\beta|^2$ for all $\alpha, \beta \in \mathbb{C}$.]

12. Let $\alpha = a+b\sqrt{-5}$ and $\beta = c+d\sqrt{-5}$. Then

$$\alpha\beta = (a+b\sqrt{-5})(c+d\sqrt{-5}) = (ac-5bd)+(ad+bc)\sqrt{-5},$$

so

$$\begin{aligned} N(\alpha\beta) &= (ac-5bd)^2+5(ad+bc)^2 \\ &= a^2c^2-10abcd+25b^2d^2+5a^2d^2+10abcd+5b^2c^2 \\ &= a^2c^2+25b^2d^2+5a^2d^2+5b^2c^2 \\ &= (a^2+5b^2)(c^2+5d^2) = N(\alpha)N(\beta). \end{aligned}$$

[Of course it also follows from the fact that $|\alpha\beta|^2 = |\alpha|^2|\beta|^2$ for all $\alpha, \beta \in \mathbb{C}$.]

13. Let $\alpha \in D$. We give a proof by induction on $|N(\alpha)|$, starting with $|N(\alpha)| = 2$, that α has a factorization into irreducibles. Let $|N(\alpha)| = 2$. Then α itself is an irreducible by Theorem 47.7, and we are done.

Suppose that every element of absolute norm > 1 but $< k$ has a factorization into irreducibles, and let $|N(\alpha)| = k$. If α is an irreducible, then we are done. Otherwise, $\alpha = \beta\gamma$ where neither β nor γ is a unit, so $|N(\beta)| > 1$ and $|N(\gamma)| > 1$. From $|N(\beta\gamma)| = |N(\beta)N(\gamma)| = |N(\alpha)| = k$, we then see that $1 < |N(\beta)| < k$ and $1 < |N(\gamma)| < k$, so by the induction assumption, both β and γ have factorizations into a product of irreducibles. The product of these two factorizations then provides a factorization of α into irreducibles.

14. Now

$$\frac{16+7i}{10-5i} = \frac{16+7i}{10-5i} \cdot \frac{10+5i}{10+5i} = \frac{125+150i}{125} = 1 + \frac{6}{5}i$$

so we let $\sigma = 1 + i$. Then $16 + 7i = (10 - 5i)(1 + i) + (1 + 2i)$. We have

$$\frac{10-5i}{1+2i} = \frac{10-5i}{1+2i} \cdot \frac{1-2i}{1-2i} = \frac{0-25i}{5} = -5i,$$

so $10 - 5i = (1 + 2i)(-5i)$. Thus $1 + 2i$ is a gcd of $16 + 7i$ and $10 + 5i$. Other possible answers are $-1 - 2i, -2 + i$, and $2 - i$.

15. a. Let $\gamma + \langle\alpha\rangle$ be a coset of $\mathbb{Z}[i]/\langle\alpha\rangle$. By the division algorithm, $\gamma = \alpha\sigma + \rho$ where either $\rho = 0$ or $N(\rho) < N(\alpha)$. Then $\gamma + \langle\alpha\rangle = (\rho + \sigma\alpha) + \langle\alpha\rangle$. Now $\sigma\alpha \in \langle\alpha\rangle$, so $\gamma + \langle\alpha\rangle = \rho + \langle\alpha\rangle$. Thus every coset of $\langle\alpha\rangle$ contains a representative of norm less than $N(\alpha)$. Because there are only a finite number of elements of $\mathbb{Z}[i]$ having norm less than $N(\alpha)$, we see that $\mathbb{Z}[i]/\langle\alpha\rangle$ is a finite ring.

b. Let π be an irreducible in $\mathbb{Z}[i]$, and let $\langle\mu\rangle$ be an ideal in $\mathbb{Z}[i]$ such that $\langle\pi\rangle \subseteq \langle\mu\rangle$. (Remember that $\mathbb{Z}[i]$ is a PID so every ideal is principal.) Then $\pi \in \langle\mu\rangle$ so $\pi = \mu\beta$. Because π is an irreducible, either μ is a unit, in which case $\langle\mu\rangle = \mathbb{Z}[i]$, or β is a unit, in which case $\mu = \pi\beta^{-1}$ so $\mu \in \langle\pi\rangle$ and $\langle\mu\rangle = \langle\pi\rangle$. We have shown that $\langle\pi\rangle$ is a maximal ideal of $\mathbb{Z}[i]$, so $\mathbb{Z}[i]/\langle\pi\rangle$ is a field.

c. i. Because $\langle 3\rangle$ contains both 3 and $3i$, we see that each coset contains a unique representative of the form $a + bi$ where a and b are both in the set $\{0, 1, 2\}$. Thus there are 9 elements in all, and the ring has characteristic 3 because $1 + 1 + 1 = 0$.

ii. By Part(**a**), each coset contains a representative of norm less that $N(1 + i) = 2$. The only nonzero elements of $\mathbb{Z}[i]$ of norm less than 2 are ± 1 and $\pm i$. Because $i = -1 + (1 + i)$ and $-i = 1 - (1 + i)$, we can reduce our list of possible nonzero cosets to $1 + \langle 1 + i\rangle$ and $-1 + \langle 1 + i\rangle$. But $[1 + \langle 1 + i\rangle] - [-1 + \langle 1 + i\rangle] = 2 + \langle 1 + i\rangle$, and $2 = (1 + i)(1 - i)$ is in $\langle 1 + i\rangle$. Thus the only cosets are $\langle 1 + i\rangle$ and $1 + \langle 1 + i\rangle$, so the order of the ring is 2, and the characteristic is 2.

iii. By Part(**a**), each coset contains a representative of norm less than $N(1 + 2i) = 5$. The only nonzero elements of $\mathbb{Z}[i]$ of norm less than 5 are of the form $a + bi$ where a and b are in the set $\{0, 1, -1\}$ or where one of a and b is ± 2 and the other is zero. These elements are $1, -1, i, -i$, $1 + i, 1 - i, -1 + i, -1 - i, 2, -2, 2i$, and $-2i$. Because

$$\begin{aligned} i &= 2 + (1 + 2i)i, \\ -i &= -2 + (1 + 2i)(-i), \\ 2i &= -1 + (1 + 2i), \\ -2i &= 1 + (1 + 2i)(-1), \\ 1 + i &= -2 + (1 + 2i)(1 - i), \end{aligned}$$

$$\begin{aligned} 1-i &= -1+(1+2i)(-i), \\ -1+i &= 1+(1+2i)(i), \\ -1-i &= 2+(1+2i)(-1+i), \end{aligned}$$

we see that every coset contains either 0, 1, -1, 2, or -2 as a representative. The ring has 5 elements and characteristic 5.

16. a. *Property 1:* Because $n > 0$, we see that $a^2 + nb^2 = 0$ if and only if $a = b = 0$.

Property 2: Let $\alpha = a + b\sqrt{-n}$ and $\beta = c + d\sqrt{-n}$. Then

$$\begin{aligned} N(\alpha\beta) &= N((ac - bdn) + (ad + bc)\sqrt{-n}) \\ &= (ac - bdn)^2 + n(ad + bc)^2 \\ &= a^2c^2 - 2abcdn + b^2d^2n^2 + a^2d^2n + 2abcdn + b^2c^2n \\ &= a^2c^2 + b^2d^2n^2 + a^2d^2n + b^2c^2n \\ &= (a^2 + nb^2)(c^2 + nd^2) = N(\alpha)N(\beta). \end{aligned}$$

[Of course it also follows from the fact that $|\alpha\beta|^2 = |\alpha|^2|\beta|^2$ for all $\alpha, \beta \in \mathbb{C}$.]

b. By Theorem 47.7, if $\alpha \in \mathbb{Z}[\sqrt{-n}]$ is a unit, then $N(\alpha) = 1$. Conversely, suppose that $N(\alpha) = 1$. Now $a^2 + nb^2 = 1$ where $n \in \mathbb{Z}^+$ if and only if either $a = \pm 1$ and $b = 0$, or $a = 0$ and $n = 1$ and $b = \pm 1$. In the former case, $\alpha = \pm 1$, and of course 1 and -1 are units. In the latter case with $n = 1$, we are in the Gaussian integers which are a Euclidean domain, and Theorem 46.6 tells us that the elements of norm 1 are indeed units.

c. The preceding parts show that we have a multiplicative norm on $\mathbb{Z}[\sqrt{-n}]$ such that the elements of norm 1 are precisely the units. By Exercise 13, every nonzero nonunit has a factorization into irreducibles.

Note that the hypothesis that n is square free was not used in this exercise. Because $a + b\sqrt{-m^2n} = a + (bm)\sqrt{-n}$, we see that the square-free assumption is really no loss of generality. The assumption that $n > 0$ was used in both Part(**a**) and Part(**b**). The square-free assumption is used in the following exercise, however.

17. a. *Property 1:* If $a^2 - nb^2 = 0$, then $a^2 = nb^2$. If $b = 0$, then $a = 0$. If $b \neq 0$, then $n = (a/b)^2$, contradicting the hypothesis that n is square free. Thus $a = 0$ and $b = 0$.

Property 2: Let $\alpha = a + b\sqrt{n}$ and $b = c + d\sqrt{n}$. Then

$$\begin{aligned} N(\alpha\beta) &= N((ac + bdn) + (ad + bc)\sqrt{n}) \\ &= (ac + bdn)^2 - n(ad + bc)^2 \\ &= a^2c^2 + 2abcdn + b^2d^2n^2 - a^2d^2n - 2abcdn - b^2c^2n \\ &= a^2c^2 + b^2d^2n^2 - a^2d^2n - b^2c^2n \\ &= (a^2 - nb^2)(c^2 - nd^2) \\ &= (a^2 - nb^2)(c^2 - nd^2) = N(\alpha)N(\beta). \end{aligned}$$

b. As an integral domain with a multiplicative norm, the norm of every unit is ± 1 by Theorem 47.7. Now suppose that $\alpha = a + b\sqrt{n}$ has norm ± 1, so that $a^2 - nb^2 = \pm 1$. Then

$$\frac{1}{\alpha} = \frac{1}{a + b\sqrt{n}} \cdot \frac{a - b\sqrt{n}}{a - b\sqrt{n}} = \frac{a - b\sqrt{n}}{a^2 - nb^2} = \pm(a - b\sqrt{n})$$